



ADVANTEST CORPORATION

SmartShell for SmarTest 7

SmartShell Client GUI

2020-11-25

Notices

Legal Notices

All rights reserved. All text and figures included in this publication are the exclusive property of Advantest Corporation. Reproduction of this publication in any manner without the written permission of Advantest Corporation is prohibited. Information in this document is subject to change without notice.

Trademarks and Registered Trademarks

- ADVANTEST is a trademark of Advantest Corporation.
- All other marks referenced herein are trademarks or registered trademarks of their respective owners.

Warranty

The material contained in this document is provided "as is," and is subject to being changed, without notice, in future editions. Further, to the maximum extent permitted by applicable law, Advantest disclaims all warranties, either express or implied, with regard to this material and any information contained herein, including but not limited to the implied warranties of merchantability and fitness for a particular purpose. Advantest shall not be liable for errors or for incidental or consequential damages in connection with the furnishing, use, or performance of this material or of any information contained herein. Should Advantest and the user have a separate written agreement with warranty terms covering the material in this document that conflict with these terms, the warranty terms in the separate agreement shall control.

Technology Licenses

The hardware and/or software described in this document are furnished under a license and may be used or copied only in accordance with the terms of such license.

Restricted Rights Legend

If software is for use in the performance of a U.S. Government prime contract or subcontract, Software is delivered and licensed as "Commercial computer software" as defined in DFAR 252.227-7014 (June 1995), or as a "commercial item" as defined in FAR 2.101(a) or as "Restricted computer software" as defined in FAR 52.227-19 (June 1987) or any equivalent agency regulation or contract clause. Use, duplication or disclosure of Software is subject to Advantest's standard commercial license terms for 93000 systems and non-DOD Departments and Agencies of the U.S. Government will receive no greater than Restricted Rights as defined in FAR 52.227-19(c)(1-2) (June 1987). U.S. Government users will receive no greater than Limited Rights as defined in FAR 52.227-14 (June 1987) or DFAR 252.227-7015 (b)(2) (November 1995), as applicable in any technical data.

Safety Notices

CAUTION

A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

WARNING

A WARNING notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in death or severe injury. Do not proceed beyond a WARNING notice until the indicated conditions are fully understood and met.

Table of Contents

1	Introduction to SmartShell Client GUI	4
2	Overview of SmartShell Client GUI	5
2.1	Config View	5
2.2	SmartShell Console View	5
2.3	Hypervisor View	6
2.4	Pattern Conversion View	6
2.5	Shmoo View	6
2.6	Event Log Console View	6
3	Config View	7
3.1	Procedure	7
4	SmartShell Console View	8
4.1	Procedure	8
5	Hypervisor View	9
5.1	Procedure	9
6	Pattern Conversion View	10
6.1	Procedure for Stil Pattern	10
6.2	Procedure for Spreadsheet based Test Program Conversion	11
7	Shmoo View	12
7.1	Procedure	12

1 Introduction to SmartShell Client GUI

The SmartShell Client GUI can be run anywhere in the network to interactively control V93000 SmarTest via the SmartShell command set. For details on SmartShell and how to install it please refer to the SmartShell reference documentation:

- 1) SmartShell for SmarTest 7 - User's Guide
- 2) SmartShell for SmarTest 7 - Command Reference.

The SmartShell Client GUI provides an easy interface:

- to connect to an active SmartShell server running on a V93000 SmarTest tester workstation;
- to interactively control SmarTest with the SmartShell command set;
- to manage multiple test tasks of any programming language for execution in a queue;
- to convert patterns in any pattern format by using a customer-specified conversion tool;
- to add setups incrementally and dynamically to the SmarTest system for a test execution;
- to perform comprehensive shmoo for an entire testflow or testsuites or pattern.

Before the user start:

The computer which runs the SmartShell Client GUI must have a java version higher than or equal to 1.7. To find it out, issue the following command in the Unix computer:

```
java -version
```

Return

```
java version "1.8.0_66"  
Java(TM) SE Runtime Environment (build 1.8.0_66-b17)  
Java HotSpot(TM) 64-Bit Server VM (build 25.66-b17, mixed mode)  
java -version
```

Note:

If it is lower than 1.8, you could either install JDK1.8, or Untar the pre-prepared JDK1.8 package(jre.tar.gz) to the folder /opt/hp93000/smartshell/gui/

To launch SmartShell Client GUI, execute the following command:

```
/opt/hp93000/smartshell/gui/ssclient &
```

2 Overview of SmartShell Client GUI

In SmartShell Client GUI window, the user can see Configuration view, SmartShell Console view, Hypervisor view, Pattern Conversion view, Shmoo view, Console view,

After the user clicks any view tab, the main window appears.

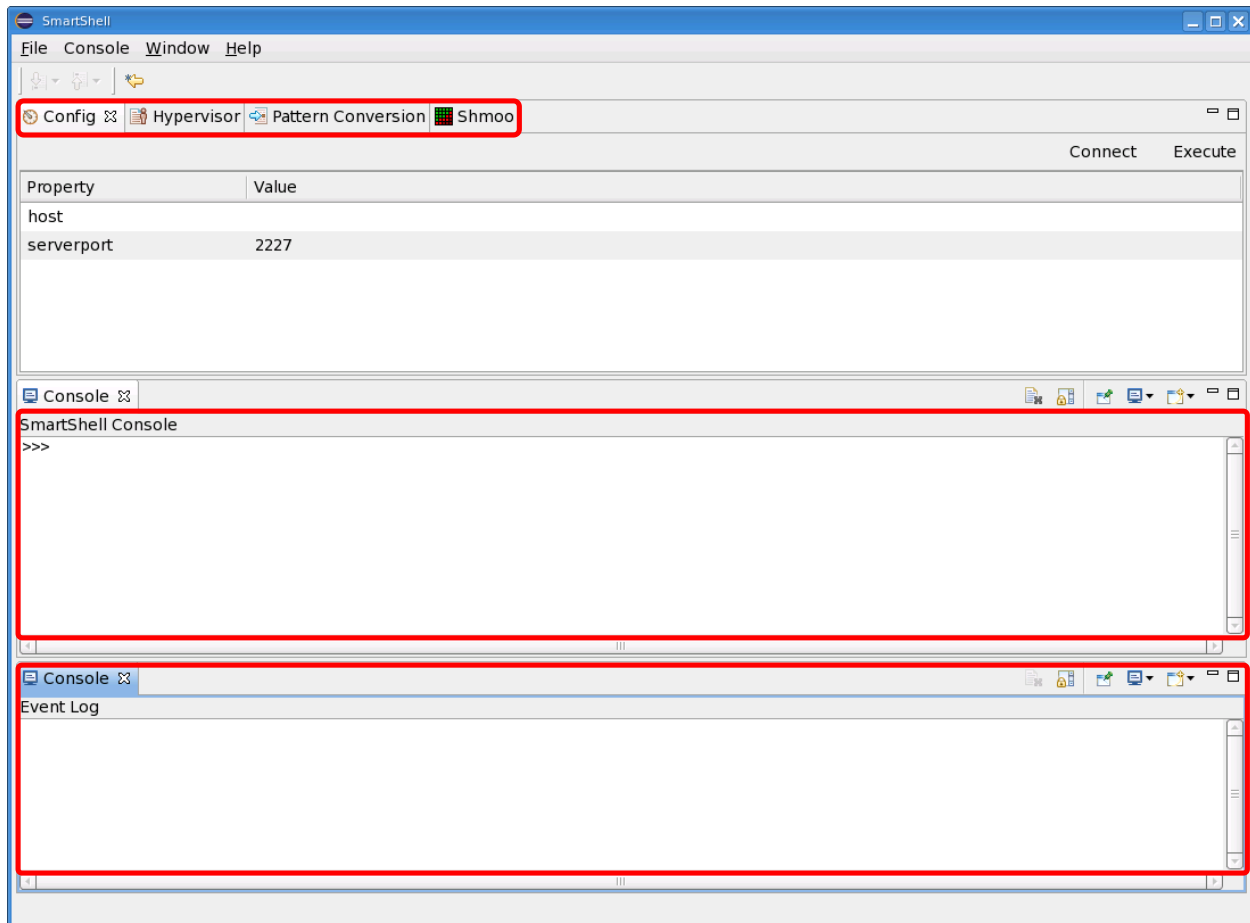


Fig. 1: SmartSheull Client GUI Window

2.1 Config View

This menu is to configure the host and server port of the SmartShell server to which the SmartShell Client GUI connects.

2.2 SmartShell Console View

In this Console, the user can interactively issue SmartShell commands to the SmartShell server for the SmartShell operations.

2.3 Hypervisor View

A task scheduler to add the user-defined test programs or test scripts to be executed in the order listed. Optionally the user can put the email address to inform the owner of each test task when the execution is completed.

2.4 Pattern Conversion View

For pattern conversion. Currently we support STIL format conversion using stilreader from TestInsight.

2.5 Shmoo View

In Shmoo view, the user can specify tester resources such as specification variables or hardware resources for the device characterization and debug.

2.6 Event Log Console View

Show the execution messages.

3 Config View

Config "host" and "serverport", then click button "connect" to connect to SmartShell server.

3.1 Procedure

- 1) Open SmartShell Client GUI and click Config tab;
- 2) Setup host and serverport, host can be empty if runs at the same workstation as SmartShell server;
- 3) Click "Connect" to connect to SmartShell Server.

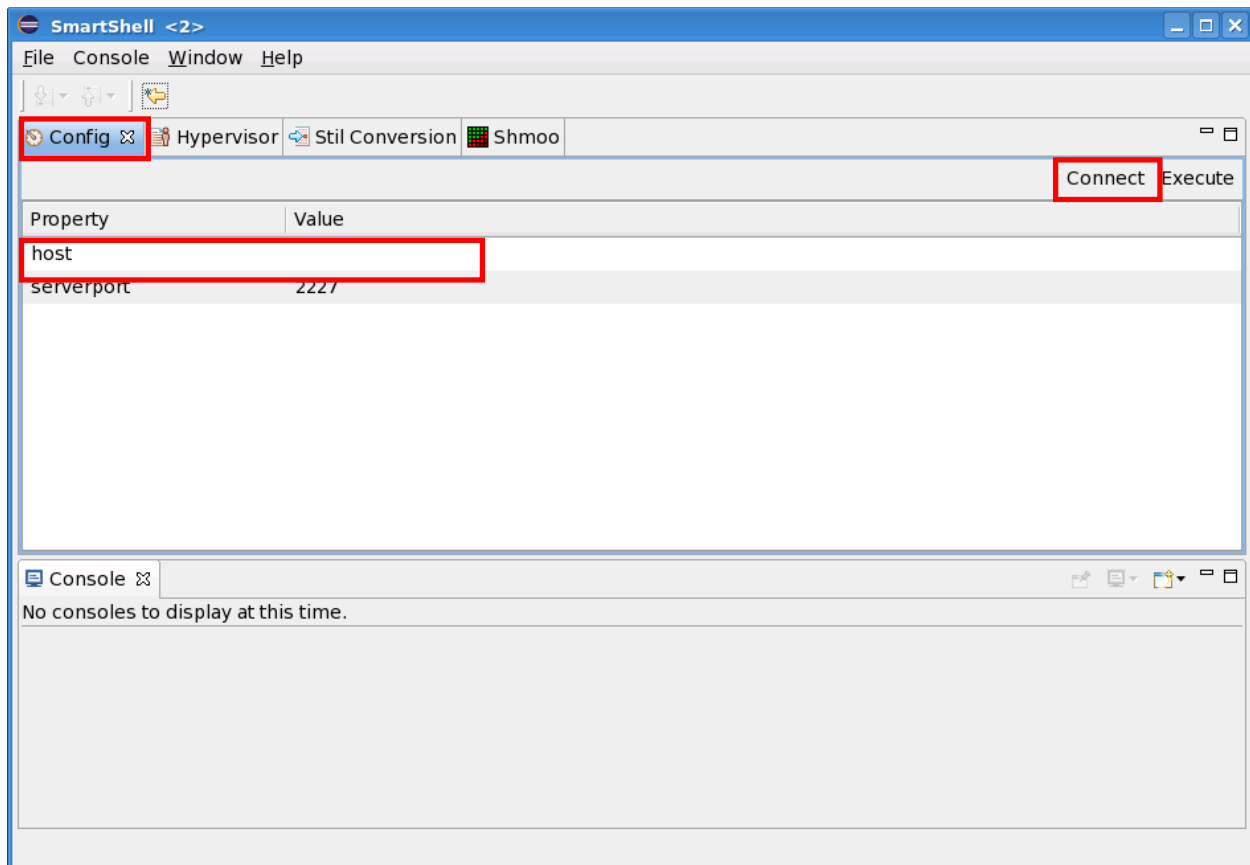


Fig. 2: SmartSheull Client GUI Window – Config View

5 Hypervisor View

Hypervisor is a SmartShell Client GUI feature that is used to schedule and execute user's test program or test script in the order listed, then to send the execution status to the relevant owner of each task.

5.1 Procedure

- 1) Open SmartShell Client GUI and click Config tab;
- 2) Setup host and serverport;
- 3) Click button "Connect" to connect to SmartShell Server;
- 4) Click Hypervisor tab;
- 5) Click button "Add" to add the scripts to excute;
- 6) Choose the script (see notes below) that user want to execute;
- 7) Click button "Execute".

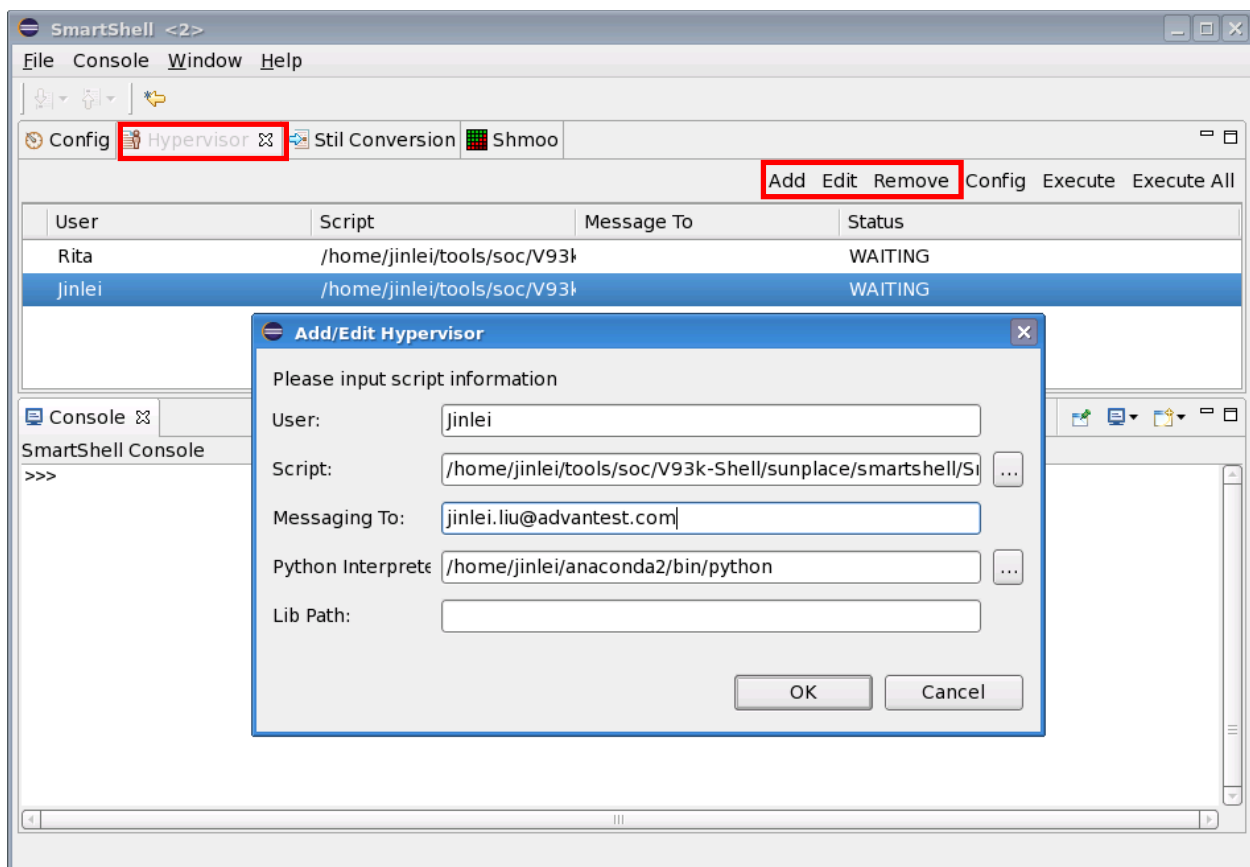


Fig. 4: SmartSheull Client GUI Window – Hypervisor View

Note:

The script can be written in any scripting language such as Python, Tcl, or perl. It must be self-contained which means this script can be executed with any other dependencies the SmartShell Client GUI needs to take care of.

6 Pattern Conversion View

Pattern Conversion is a SmartShell Client GUI feature that can be used to convert pattern such as stil patterns, then execute the generated burst in SmarTest. This pattern conversion GUI supports *stilreader* for stil pattern conversion and also *convmgr* for spreadsheet based test program conversion.

6.1 Procedure for Stil Pattern

- 1) Open SmartShell Client GUI and click Config tab;
- 2) Setup host and serverport;
- 3) Click button "Connect" to connect to SmartShell Server;
- 4) Click Pattern Conversion tab;
- 5) Select STIL Conversion
- 6) Choose the conversion setup file in "Conversion Setup";
- 7) Choose the Pattern file such as stil in "Pattern file";
- 8) Click button "Run";
- 9) Click "Run Pattern Shmoo" to go to the Shmoo View.

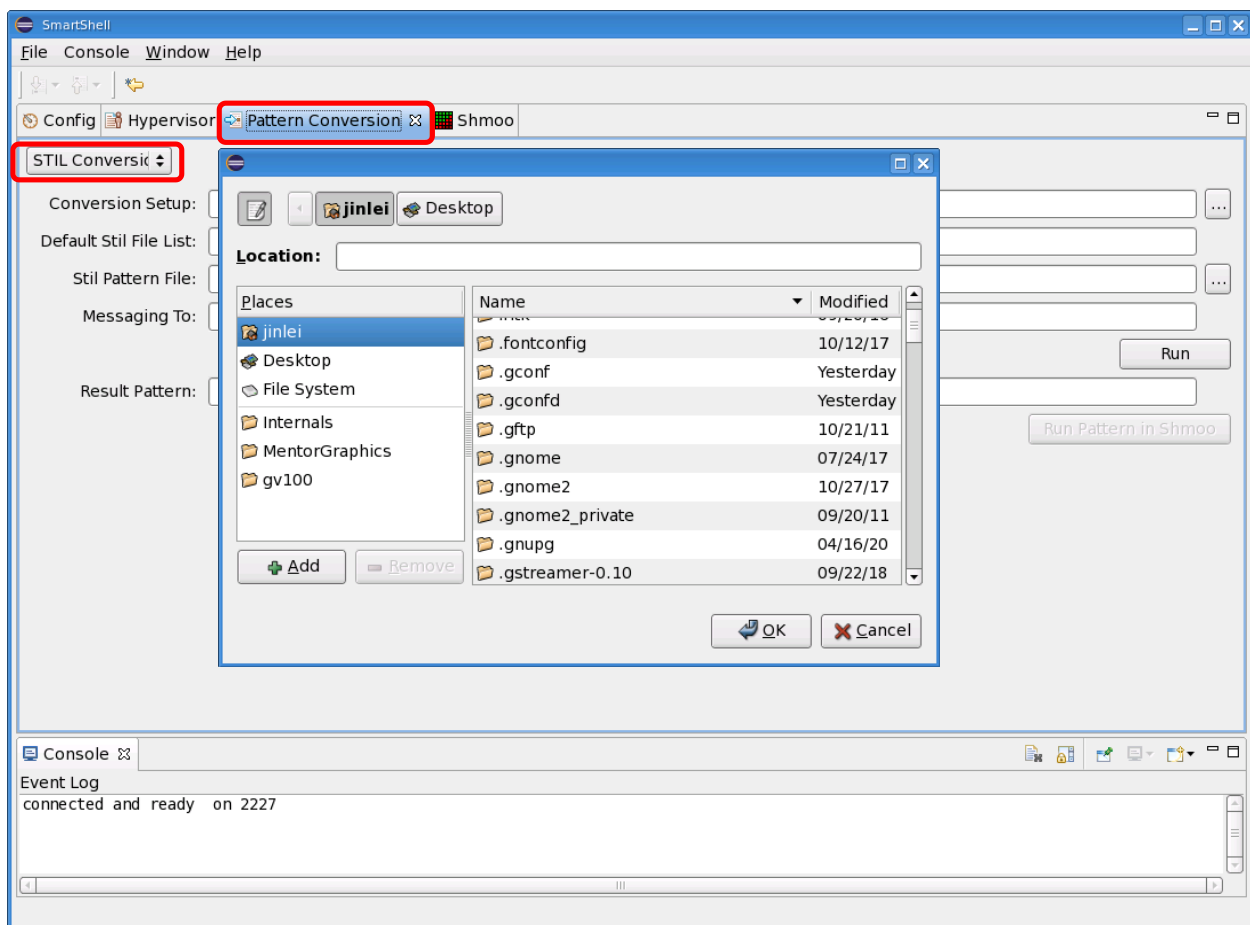


Fig. 5: SmartSheull Client GUI Window – Pattern Conversion View

6.2 Procedure for Spreadsheet based Test Program Conversion

- 1) Open SmartShell Client GUI and click Config tab;
- 2) Setup host and serverport;
- 3) Click button "Connect" to connect to SmartShell Server;
- 4) Click Pattern Conversion tab;
- 5) Select ConvManager
- 6) Choose the conversion setup file in "Spread File";
- 7) Click button "Run";
- 8) Click "Shmoo" to go to the Shmoo View.

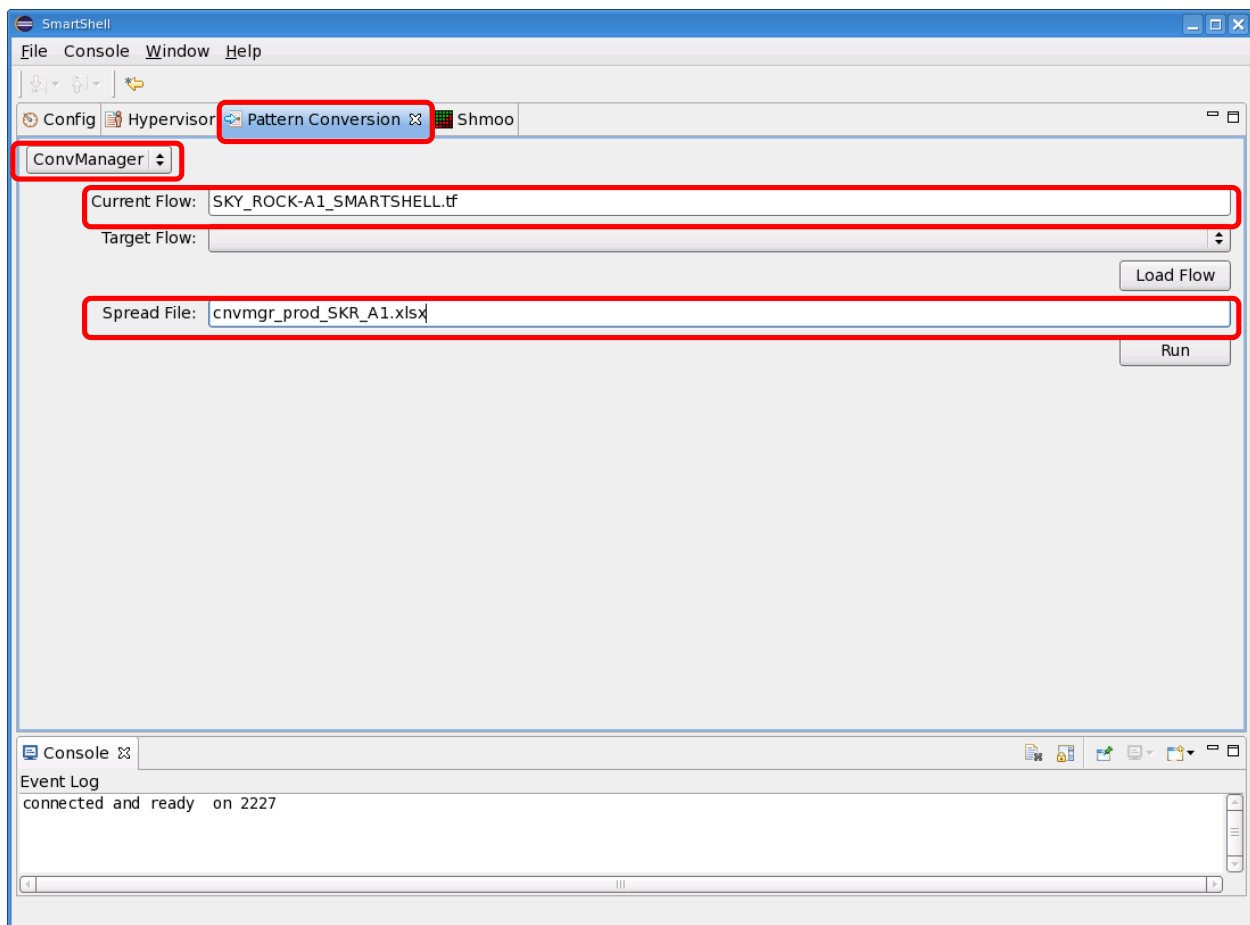


Fig. 6: SmartSheull Client GUI Window – Pattern Conversion View

The converted patterns will be added to the test program and new testsuites will be added to a temporary testflow shown as __tmp_<testflow> in the SmarTest Work Center.

If the new pattern has the same name as the existing pattern, the existing pattern will be replaced. If the new testsuite has the same name as those in the current testflow, the testsuite will be appended to the testflow with an index number as <testsuite>_n where n will be incremented.

7 Shmoo View

Shmoo is a SmartShell Client GUI feature that do shmoo debug with the specified spec variables.

7.1 Procedure

- 1) Open SmartShell Client GUI and click Config tab;
- 2) Setup host and serverport;
- 3) Click button "Connect" to connect to SmartShell Server;
- 4) Click Shmoo tab;
- 5) Click "Refresh" button;
- 6) Select Execution Type: Pattern, Flow, Suite;
- 7) Select Pattern, or Flow or Suite;
- 8) Select timing specification;
- 9) Select level specification;
- 10) Define the spec variables in the setup table by clicking the x-axis and y-axis;
- 11) Click button "Run".

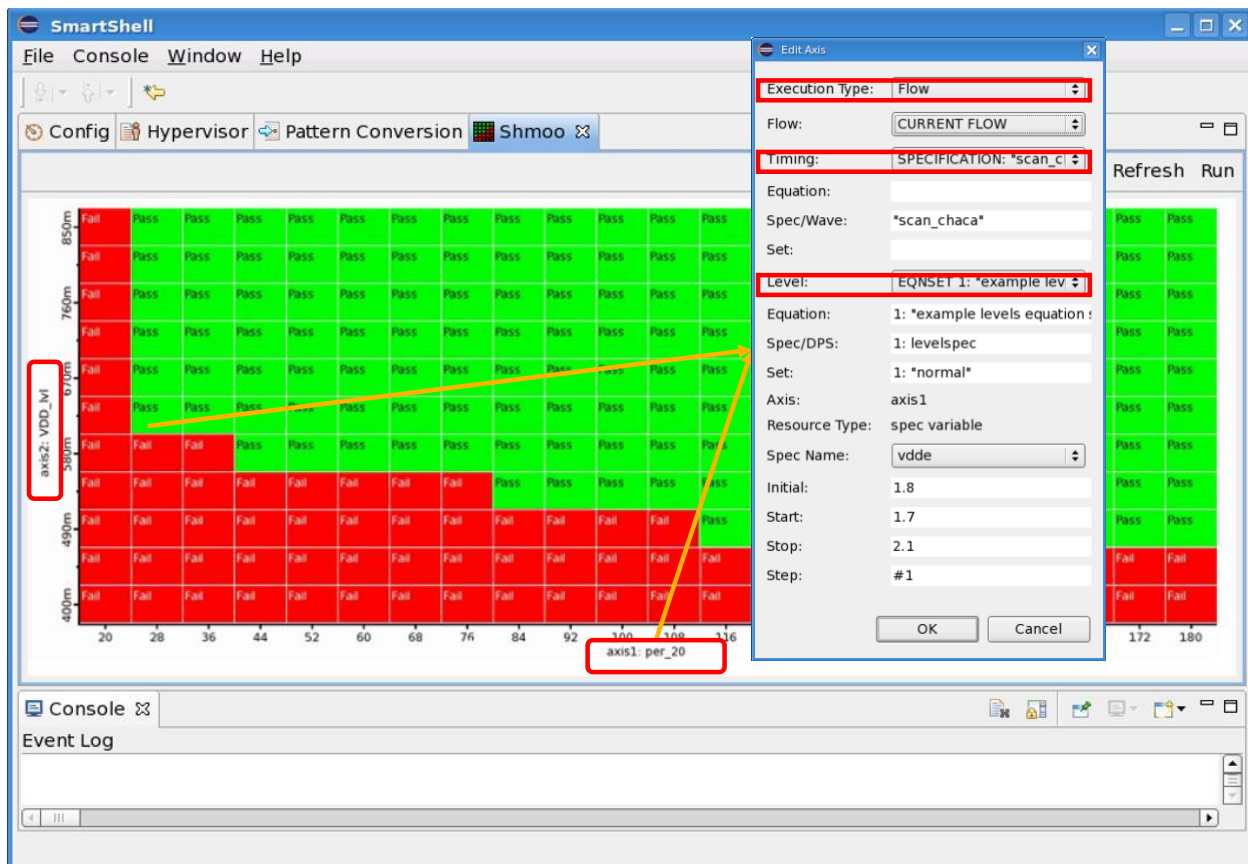


Fig. 7: SmartSheull Client GUI Window – Shmoo View